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Some of our Favorite Functions in the complex plane.

We have already looked at e^z and $\log(z)$ and seen how they are a bit different in the complex plane. Let's investigate some other functions we know from the real numbers in the same way.

The Power Function $f(z) = z^c$

We define the power function for complex z, c in the following way:

$$z^c = e^{c\,log(z)}$$

To see why this works let's take c as an integer, something we have dealt with.

Suppose $z = re^{i\theta}$ then $z^c = r^c e^{ic\theta}$ and with the definition $z^c = e^{clog(z)} = \left(e^{\log(z)}\right)^c = \left(re^{i\theta}\right)^c = r^c e^{ic\theta}$

The fact that this definition coincides when c is an integer or $c = \frac{1}{n}$ is what motivates this definition of z^c for any complex c

We can also say that $\frac{1}{z^c} = \frac{1}{e^{clog(z)}} = e^{-c\log(z)} = z^{-c}$

This function is however multi-valued but we can make it single valued by choosing some branch cut $\alpha < \theta < \alpha + 2\pi$ For $\log(z) = \ln(r) + i\theta$. If we do this we then can look at the derivative.

$$\frac{\mathrm{d}}{dz}z^c = \frac{\mathrm{d}}{dz}\mathrm{e}^{\mathrm{c}\log(\mathrm{z})} = \frac{\mathrm{c}}{\mathrm{z}}\mathrm{e}^{\mathrm{clog(z)}}$$

You might think, wait, you said $\frac{d}{dz}z^c = cz^{c-1}$ well indeed it does, recall again that $z = e^{\log(z)}$ let's plug this into the result above and verify that they indeed are the same!

We say that we take the Principal value(P.V.) of z^c when we use $Log(z)$ instead of $\log(z)$

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$z^c = e^{c\log(z)}$ $\quad c^z$

The exponential function of base *c*

From our definition of the ~~power~~ Power function we can define the exponential function of base c by simply switching z and c!

$$c^z = e^{z\,log(c)}$$

Let's take the derivative as before:

$$\frac{d}{dz}c^z = \frac{d}{dz}e^{z\log(c)} = \log(c)e^{z\log(c)} = \log(c)c^z$$

Homework Example:

$\sqrt{2}^i =$

1. Compute the following (show work)
   (a) $(1+i)^i$
   (b) $(-1)^{\frac{1}{\pi}}$

c^c $\quad c\log(z)$

$z = e$

$1+i$

a) $(1+i)^i$

$= e^{i\log(1+i)}$

$= e^{i\log(\sqrt{2}e^{i\pi/4})}$

$= e^{i(\ln(\sqrt{2}) + i(\pi/4 + 2n\pi))}$

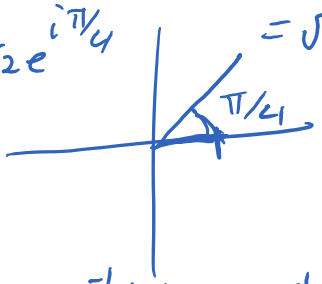
$= e^{i\ln(\sqrt{2})} e^{-\pi/4 - 2n\pi}$

$(1+i)^i = e^{-\pi/2 - 2n\pi} e^{i\ln(\sqrt{2})} = re^{i\theta}$

$\log(z) = \ln|r| + i\theta$

$1+i = re^{i\theta}$ $\quad r = \sqrt{1^2+1^2} = \sqrt{2}$

$1+i = \sqrt{2}e^{i\pi/4}$



$\tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{1}{1}\right)$

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Rules for exponents

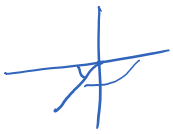
$z^c = e^{c\log(z)}$

While we do have some familiar rules for exponents in the complex number system, namely:

1. $z^c z^d = z^{c+d}$
2. $\frac{z^c}{z^d} = z^{c-d}$
3. $(z^c)^n = z^{cn}$
4. $\frac{1}{z^c} = z^{-c}$ *n an integer.*

Others won't for certain numbers.

$z_1 = \sqrt{2}e^{i\pi/4}$ $z_2 = \sqrt{2}e^{-i3\pi/4}$



Let $z_1 = 1 + i$, $z_2 = -1 - i$ Let's show by taking the principal values that $(z_1 z_2)^i \neq z_1^i z_2^i$

$(z_1 z_2)^i = \left(\sqrt{2}e^{i\pi/4}\sqrt{2}e^{-i3\pi/4}\right)^i$

$= \left(2e^{i(\pi/4 - 3\pi/4)}\right)^i$

$= \left(2e^{i(-\pi/2)}\right)^i$

$= e^{i\log(2e^{-i\pi/2})}$

$= e^{i(\ln(2) + -\pi/2 i)}$

$= e^{i\ln(2) + \pi/2}$

$= e^{\pi/2} e^{i\ln(2)}$

$z_1^i z_2^i = \left(\sqrt{2}e^{i\pi/4}\right)^i \left(\sqrt{2}e^{-i3\pi/4}\right)^i$

$= e^{i\log(\sqrt{2}e^{i\pi/4})} e^{i\log(\sqrt{2}e^{i(-3\pi/4)})}$

$= \left(e^{i(\ln\sqrt{2} + i\pi/4)} e^{i(\ln(\sqrt{2}) - i3\pi/4)}\right)$

$= e^{i\ln(\sqrt{2})} e^{-\pi/4} e^{i\ln(\sqrt{2})} e^{+3\pi/4}$

$e^{i\ln\sqrt{2}} e^{i\ln(\sqrt{2})} e^{-\pi/4} e^{3\pi/4}$

$e^{i(\ln(\sqrt{2}) + \ln(\sqrt{2}))} e^{-\pi/4 + 3\pi/4}$

$e^{i\ln(\sqrt{2}^2)} e^{\pi/2}$

$e^{i\ln(2)} e^{\pi/2}$

$\ln(AB) = \ln A + \ln B$

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The Trigonometric Functions

We know that $e^{iz} = \cos(z) + i\sin(z)$ it turns out that we can use this to define the trig functions of a complex variable, that is

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2} \qquad \sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

If is a real number these formulas correspond exactly to the real versions of sine and cosine.

Let's just check that let $z = x$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$= \frac{\cos(x) + i\sin(x) + \cos(-x) + i\sin(-x)}{2}$$

$$= \frac{\cos(x) + i\sin(x) + \cos(x) - i\sin(x)}{2} = \frac{2\cos(x)}{2} = \cos(x)$$

$$\sin(x) = \frac{\cos(x) + i\sin(x) - (\cos(x) - i\sin(x))}{2i}$$

$$= \frac{\cos(x) + i\sin(x) - \cos(x) + i\sin(x)}{2i}$$

$$\frac{2i\sin(x)}{2i} = \sin(x)$$

Further since e^z is entire so are $\cos(z)$ and $\sin(z)$! Let's find their derivatives using the definition

$$\cos(z) = \frac{e^{iz} + e^{-iz}}{2} \qquad \sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$$

$$\frac{d}{dz}\cos(z) = \frac{d}{dz}\frac{e^{iz} + e^{-iz}}{2} = \frac{\frac{d}{dz}e^{iz} + \frac{d}{dz}e^{-iz}}{2}$$

$$= \frac{ie^{iz} - ie^{-iz}}{2} = \frac{i(e^{iz} - e^{-iz})}{2}$$

$$\frac{1}{i} = -i$$

$$= -\frac{(e^{iz} - e^{-iz})}{2i} = -\sin(z)$$

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We also get some of our favorite trig identities as well.

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos(z_1)\sin(z_2)$$
$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin(z_1)\sin(z_2)$$

We can use these to show

$$\sin(2z) = 2\sin(z)\cos(z), \cos(2z) = \cos^2 z - \sin^2 z \ , \sin\left(z + \frac{\pi}{2}\right) = \cos(z), \sin\left(z - \frac{\pi}{2}\right) = -\cos(z)$$

The Pythagorean identity holds as well $\sin^2 z + \cos^2 z = 1$

The functions also keep their 2π Periodicity as adding 2π amounts to multiplying by 1, Let's just show that.

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One thing we have been doing is decomposing our complex functions into forms like $f(z) = u(x,y) + iv(x,y)$. Naturally we might ask, what is this decomposition for our trig functions.

$$\sin(z) = \sin(x)\cosh(y) + i\cos(x)\sinh(y)$$

$$\cos(z) = \cos(x)\cosh(y) + i\sin(x)\sinh(y)$$

Let's show this for $\sin(z)$ recall that $\cos(y) = \frac{e^y + e^{-y}}{2}$, $\sinh(y) = \frac{e^y - e^{-y}}{2}$ and we will employ our additive identity with $z_1 = x \;\; z_2 = iy$ noting that $\sin(iy) = i\sinh(y)$ and $\cos(\mathrm{iy}) = \cosh(\mathrm{y})$

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As a result of these formulas we can show one major difference between sine and cosine on the real line vs the complex plane. On the real line we know that $|\sin(x)| < 1$ and $|\cos(x)| < 1$ However in the complex plane this is not true…

$$\sin(z) = \sin(x)\cosh(y) + i\cos(x)\sinh(y)$$

$$\cos(z) = \cos(x)\cosh(y) + i\sin(x)\sinh(y)$$

Let's take the moduli and show this for sine at least.

$$|\sin(z)|^2 = \sin^2(x) + \sinh^2(y)$$

$$|\cos(z)|^2 = \cos^2(x) + \sinh^2(y)$$

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As it turns out unlike the case of some functions, like $x^2 + 1$, extending the sine and cosine functions into the complex plane does not change their zeros! In fact we have the following theorem.

Theorem: The zeros of $\sin(z)$ and $\cos(z)$ are the same in the complex plane as the zeros of $\sin(x)$ and $\cos(x)$ on the real line. That is:

$\sin(z) = 0$ if and only if $z = n\pi$ where n is an integer

And

$\cos(z) = 0$ if and only if $z = \frac{\pi}{2} + n\pi$ where n is an integer

It is easy to show this, let's do it for $\sin(z)$

Recall $|\sin(z)|^2 = \sin^2(x) + \sinh^2(y)$

Let's suppose $\sin(z) = 0$ and see what that tells us.

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What about the other Trig functions? It turns out that they are define in the usual way .

$$\tan(z) = \frac{\sin(z)}{\cos(z)} \qquad \cot(z) = \frac{\cos(z)}{\sin(z)}$$

$$\sec(z) = \frac{1}{\cos(z)} \qquad \csc(z) = \frac{1}{\sin(z)}$$

One can also show that they have the usual derivatives

$$\frac{d}{dz}\tan(z) = \sec^2(z) \qquad \frac{d}{dz}\cot(z) = -\csc^2(z)$$

$$\frac{d}{dz}\sec(z) = \sec(z)\tan(z) \qquad \frac{d}{dz}\csc(z) = -\csc(z)\cot(z)$$

Homework Example

4. Show that

(a) $\sinh(z + i\pi) = -\sinh(z)$

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Hyperbolic Functions in the complex plane

The Hyperbolic trig functions retain their definition in the complex plane that is.

$$\sinh(z) = \frac{e^z - e^{-z}}{2} \qquad \cos(z) = \frac{e^z - e^{-z}}{2}$$

Their derivatives are also the same

$$\frac{d}{dz}\sinh(z) = \cosh(z) \qquad \frac{d}{dz}\cosh(z) = \sinh(z)$$

In contrast to the regular trig functions the zeros of the hyperbolic trig functions lie on the imaginary axis instead of the real line.

Theorem: The zeros of $\sinh(z)$ and $\cosh(z)$ line in the complex plane lie on the imaginary axis and ,

$\sinh(z) = 0$ if and only if $z = n\pi i$

And

$\cosh(0) = 0$ if and only if $z = \left(\frac{\pi}{2} + n\pi\right) i$

Where n is an integer.

You might be realizing how related trig and hyperbolic trig functions are, after they are all definable with the exponential! Even their derivative structure is very similar.

$$\frac{d}{dz}\tanh(z) = \sec^2(z) \qquad \frac{d}{dz}(\coth(z) = -\operatorname{csch}^2(z)$$

$$\frac{d}{dz}\operatorname{sech}(z) = -\operatorname{sech}(z)\tanh(z) \qquad \frac{d}{dz}\operatorname{csch}(z) = -\operatorname{csch}(z)\coth(z)$$

Where naturally,

$$\tanh(z) = \frac{\sinh(z)}{\cosh(z)} \quad , \quad \operatorname{csch}(z) = \frac{1}{\sinh(z)}, \quad \operatorname{sech}(z) = \frac{1}{\cosh(z)}, \quad \coth(z) = \frac{\cosh(z)}{\sinh(z)}$$

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Inverses of trig and hyperbolic trig functions

We know how useful $\arcsin(x)$, $\arccos(x)$, and $\arctan(x)$ can be in calculus and we would like the same kind of function here.

Let's start with $\arcsin(z)$ if $w = \arcsin(z)$ then this must mean $z = \sin(w)$

That is $z = \frac{e^{iw} - e^{-iw}}{2}$ Let's try and solve this for w….

Then $\arcsin(z) = -ilog\left[iz + (1 - z^2)^{\frac{1}{2}}\right]$. This function is very multivalued infact it has infinately many values for a given z!

To illustrate, let's look at $\arcsin(-i) = -ilog(1 \pm \sqrt{2})$

With the plus sign, $-ilog\left(1 + \sqrt{2}\right) = -i(\ln\left(1 + \sqrt{2}\right) + 2n\pi)$ and with the minus sign $-i(\ln\left(1 - \sqrt{2}\right) + \pi + 2n\pi)$

We can play the same quadratic solution game as above to find that :

$\arccos(z) = -ilog\left[z + i(1 - z^2)^{\frac{1}{2}}\right]$ and $\arctan(z) = \frac{i}{2}\log\left(\frac{i+z}{i-z}\right)$

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The derivatives follow those for the real counterparts of the inverse trig functions that is.

$$\frac{d}{dz}\arcsin(z) = \frac{1}{\sqrt{1-z^2}} \qquad \frac{d}{dz}\arccos(z) = \frac{-1}{\sqrt{1-z^2}} \qquad \frac{d}{dz}\arctan(z) = \frac{1}{1+z^2}$$

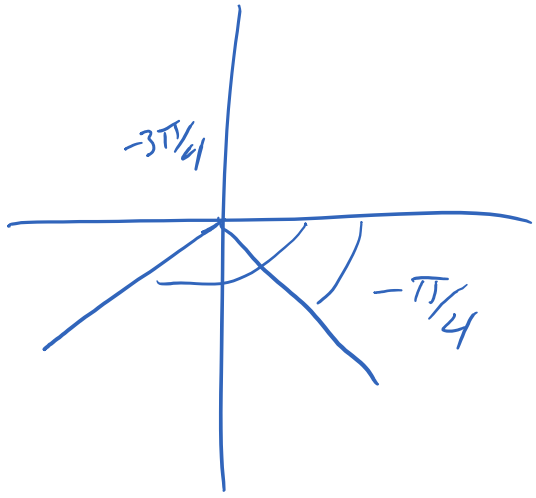
Our inverse hyperbolic functions are given by:

$$\operatorname{arcsinh}(z) = \log\left[z + (z^2+1)^{\frac{1}{2}}\right] \qquad \operatorname{arccosh}(z) = \log[z + (z^2-1)^{\frac{1}{2}}] \qquad \operatorname{arctanh}(z) = \frac{1}{2}\log\left(\frac{1+z}{1-z}\right)$$

Homework Example

5. Find all the values of
   (a) $\tan^{-1}(2i)$
   (b) $\tan^{-1}(1+i)$
   (c) $\cosh^{-1}(-1)$
   (d) $\tanh^{-1}(0)$

$z_1 = (1-i)$ $z_2 = (-1-i)$



$$(z_1 z_2)^i = \left(\sqrt{2}e^{-\pi/4 i} \sqrt{2}e^{-3\pi/4 i}\right)^i$$

$$= \left(2e^{-(4\pi/4)i}\right)^i$$

$$= (-2)^i = e^{i\log(2e^{i\pi})}$$

$$= e^{i(\ln(2)+i\pi)}$$

$$= e^{i\ln(2)-\pi}$$

$$= e^{-\pi} e^{i\ln(2)}$$

$$(1-i)^i(-1-i)^i = \left(\sqrt{2}e^{i(-\pi/4)}\right)^i \left(\sqrt{2}e^{i(-3\pi/4)}\right)^i$$

$$= e^{i(\log(\sqrt{2}e^{i(-\pi/4)}))} \; e^{i\log(\sqrt{2}e^{i(-3\pi/4)})}$$

$$= e^{i(\ln(\sqrt{2})-i\pi/4)} \; e^{i(\ln(\sqrt{2})-i3\pi/4)}$$

$$= e^{i(\ln\sqrt{2}+\ln\sqrt{2}-i(\pi/4+3\pi/4))} = e^{i\ln(2)} e^{\pi}$$